Design Documentation

Google Ads Automated Reporting Tool

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1. Executive Summary

This document describes the design and architecture of our Google Ads Automated Reporting Tool. The tool uses the Google Ads API to retrieve campaign performance data and generate automated reports and dashboards for small and medium-sized businesses. The tool operates in a read-only capacity and does not modify any campaign settings, budgets, or ad content.

2. Product Overview

2.1 Purpose

The tool automates the process of collecting, aggregating, and visualizing Google Ads campaign data. It replaces manual data exports and spreadsheet-based reporting with an automated pipeline that delivers scheduled reports to clients.

2.2 Target Users

- Small and medium-sized businesses running Google Ads campaigns
- Marketing managers who need regular performance insights
- Business owners who want transparency into their ad spend

2.3 Core Features

Feature	Description

Automated Data Retrieval	Scheduled pulls of campaign, ad group, and keyword performance data via the Google Ads API
KPI Dashboards	Visual dashboards showing key metrics: CTR, CPC, conversions, ROAS, cost-per-acquisition
Scheduled Reports	Automated weekly and monthly PDF/email reports delivered to clients
Performance Alerts	Email notifications when KPIs exceed or fall below defined thresholds
Historical Analysis	Trend analysis and period-over-period comparisons

3. Architecture

3.1 System Overview

```
+---------------------+         +----------------------+         +------------------+
|                     |         |                      |         |                  |
|   Google Ads API    | ------> |   Data Pipeline      | ------> |   Data Storage   |
|   (Read-Only)       |         |   (Cloud Functions)  |         |   (BigQuery)     |
|                     |         |                      |         |                  |
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                                |                      |         |                  |
                                |   Email Reports      | <------ |   Dashboard      |
                                |   (Scheduled)        |         |   (Looker Studio)|
                                |                      |         |                  |
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```

3.2 Technology Stack

Component	Technology
API Integration	Google Ads API v18 (REST), Python client library

Data Pipeline	Google Cloud Functions (Python 3.12)
Scheduling	Google Cloud Scheduler
Data Storage	Google BigQuery
Visualization	Looker Studio (formerly Google Data Studio)
Notifications	Email via Gmail API / Slack Webhooks
Hosting	Google Cloud Platform

4. Google Ads API Usage

4.1 Access Level

The tool requires **read-only access** to Google Ads accounts. It does not create, modify, or delete any campaigns, ad groups, ads, or keywords. No billing or account management operations are performed.

4.2 API Resources Used

Resource	Purpose	Access Type
GoogleAdsService.SearchStream	Query campaign, ad group, and keyword performance metrics	Read-only
Customer	Retrieve account information and linked accounts	Read-only
Campaign	Retrieve campaign names, status, and settings for reporting context	Read-only
AdGroup	Retrieve ad group structure for detailed breakdowns	Read-only

Metrics	Impressions, clicks, conversions, cost, CTR, CPC, ROAS	Read-only

4.3 API Query Examples

Typical GAQL query used to retrieve campaign performance data:

```
SELECT
  campaign.name,
  campaign.status,
  metrics.impressions,
  metrics.clicks,
  metrics.cost_micros,
  metrics.conversions,
  metrics.conversions_value,
  metrics.ctr,
  metrics.average_cpc,
  segments.date
FROM campaign
WHERE segments.date DURING LAST_30_DAYS
  AND campaign.status != 'REMOVED'
ORDER BY metrics.cost_micros DESC
```

4.4 Rate Limiting & Best Practices

- API calls are batched and scheduled during off-peak hours
- Data is retrieved once daily (configurable per client) to minimize API usage
- Results are cached in BigQuery to avoid redundant API calls
- Exponential backoff is implemented for transient errors
- All API usage stays well within Google's rate limits

5. Data Flow

5.1 Data Collection Process

1. **Trigger:** Cloud Scheduler triggers a Cloud Function on a daily schedule

2. **Authentication:** OAuth 2.0 refresh token is used to authenticate with the Google Ads API
3. **Data Retrieval:** GAQL queries pull performance metrics for the configured date range
4. **Transformation:** Raw API responses are normalized and enriched with calculated KPIs
5. **Storage:** Processed data is written to BigQuery tables (partitioned by date)
6. **Visualization:** Looker Studio dashboards query BigQuery in real-time
7. **Reporting:** Scheduled report jobs generate and email PDF summaries to clients

5.2 Authentication & Security

- OAuth 2.0 is used for all API authentication
- Refresh tokens are stored encrypted in Google Cloud Secret Manager
- Each client's data is isolated in separate BigQuery datasets
- Access to dashboards is controlled via Google account permissions
- No Google Ads credentials are stored in application code or version control

6. User Interaction Flow

6.1 Client Onboarding

1. Client grants read-only access to their Google Ads account via OAuth consent screen
2. Tool retrieves and stores the OAuth refresh token securely
3. Initial data backfill is performed (last 90 days)
4. Dashboard is configured and shared with the client
5. Automated reporting schedule is set up (weekly/monthly)

6.2 Ongoing Usage

- Data is automatically updated daily — no manual intervention required
- Clients access their dashboards via a shared Looker Studio link
- Performance alerts are delivered via email when thresholds are triggered
- Scheduled PDF reports are sent to the client's email

7. Compliance

- The tool complies with the **Google Ads API Terms of Service**
- Only read-only API access is used — no campaign modifications are made
- User data is processed in accordance with **GDPR** regulations
- Clients can revoke access at any time via their Google Account settings
- All data is hosted within the EU (Google Cloud region europe-west3, Frankfurt)
- A privacy policy is published at **jochenreider.com/#datenschutz**

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